

Project Development Phase - 1st PDF

Data Preparation and Business Questions with Visuals

Project Overview

The Food Ordering Behaviour and Consumer Trends dashboard was developed to analyze order demand, spending patterns, cuisine and meal-type preferences, city-wise trends, and delivery performance. The project uses Tableau to transform raw order data into interactive visualizations that support business decision-making.

Data Preparation

Before creating the dashboard, the dataset was cleaned and prepared to ensure accurate analysis.

Data Preparation Steps

- Imported the food ordering dataset into Tableau.
- Verified data types for all fields (Order ID, Order Value, Delivery Fee, Cuisine, Meal Type, City, Time Taken for Delivery).
- Removed duplicate order records.
- Handled missing or null values.
- Standardized text values (city and cuisine names) for consistency.
- Checked numerical fields (Order Value, Delivery Fee, Delivery Time) for accuracy.
- Created calculated fields for KPIs.
- Organized dimensions and measures.
- Validated data before visualization.

Key Performance Indicators (KPIs)

Total Orders

Displays the total number of food orders placed during the selected period.

Total Order Value

Represents the total revenue generated from all customer orders.

Average Delivery Time

Shows the average time taken to deliver an order, a key operational efficiency metric.

Top Cuisine / Active Restaurants

Highlights the highest-demand cuisine and the number of active restaurants on the platform.

Business Questions

Business Question 1: How many total food orders were placed?

Visualization: KPI Card | Insight: Monitors overall business activity and demand.

Business Question 2: What is the total order value generated?

Visualization: KPI Card | Insight: Measures overall sales performance.

Business Question 3: What is the average delivery time?

Visualization: KPI Card | Insight: Indicates operational efficiency and customer experience.

Business Question 4: Which cuisine is ordered the most?

Visualization: Bar Chart | Insight: Identifies the most popular cuisines and customer preferences.

Business Question 5: Which meal type is preferred by customers?

Visualization: Pie / Donut Chart | Insight: Helps understand mealtime demand and supports staffing decisions.

Business Question 6: Which cities generate the highest order demand?

Visualization: Map / Bar Chart | Insight: Reveals top-performing cities and expansion opportunities.

Business Question 7: How is delivery time distributed across orders?

Visualization: Histogram / Box Plot | Insight: Evaluates delivery consistency and identifies delay-prone segments.

Business Question 8: How does order value relate to delivery fee?

Visualization: Scatter Plot | Insight: Assesses whether delivery pricing aligns with order value.

Dashboard Features

- Interactive filters (City, Cuisine, Meal Type)
- KPI cards
- Bar charts
- Pie / donut charts
- Map visualization
- Tooltips
- Legends
- Dynamic highlighting

Benefits

- Identifies customer and cuisine preferences.
- Tracks business and delivery performance.
- Supports revenue and demand analysis.
- Improves delivery efficiency.
- Enables data-driven decision-making.
- Provides interactive, self-serve business insights.

Conclusion

The dashboard converts raw food ordering data into meaningful visual information. Through proper data preparation and interactive visualizations, the business can better understand ordering behaviour,

monitor delivery performance, and make informed decisions to improve operational efficiency and customer experience.